

INNOVUS COMMANDS

INNOVUS commands at floorplan stage:

1) **create_floorplan -box_size {die_box io_box core_box}**

The **create_floorplan** command initializes the floorplan and calls the **add_tracks** command to create new routing tracks.

2) **create_row -area {x1 y1 x2 y2}**

Create rows in the specified rectangle

3) **create_tracks -direction -layer -num -start -step**

Creates non uniform and customized tracks in the selected design area

-direction — specifies tracks direction as a string

-start —starting co-ordinate to create the track

-layer —specifies the routing layer used for the tracks, more than 1 layer can specified

-step —specifies the spacing between the tracks

-num —specifies the number of tracks to create for the grid

4) **delete_tracks -direction -layer**

Delete the set of tracks in the database for a given metal layer.

5) **edit_pin -pin_name -location -edge -layer -snap {track} -spacing**

Modifies the properties of pins, such as pin spreading, pin location, width and depth etc

6) **init_design**

To load the netlist file

7) read_def

Load a DEF file containing the floorplan saved by any tool.

Load the DEF file containing the scan chain information so that the placement can do scan chain reorder.

8) snap_floorplan {-all | -selected }

Snap floorplan command is used to snap blocks, I/O Pads, standard cells to the FINEST grid.

9) check_place

Checks FIXED and PLACED cells for violations, adds violations markers to the display area, and generates the violation report.

10) add_io_buffers

Adds buffers/inverters to the I/O pins and places the buffers/inverters near the I/O pins.

11) check_pin_assignment

Sanity checks with respect to port placement like spacing, alignment with tracks, alignment with proper metal layer etc.

12) legalize_pins -keep_layer -keep_order

This command moves a pin to the nearest legal location. Moves a pin from its existing location in the following cases: /

a) Pin overlap with any other object or pin.

b) If the pin is not in the non preferred layer, a non reserved layer, or the metal layer.

c) If the pin is not on the routing track.

d) If the pin does not honor corner mask settings.

-keep_layer — Maintains the pin layer while legalizing the pins.

-keep_order — Maintains the pin order while legalizing the pins.

13) create_route_blockage

Creates an area which prevents which prevents routing of specified metal layer, signal routes and hierarchical instances.

{-insts name } - blockage over a particular instance

{-layers layer name } -blockage over particular layer

{-rects {x1 y1 x2 y2} ... { }} specifies the rectangular area of the blockage

{polygon {x1 y1 x2 y2}...{ } }-specifies the polygon area of the blockage
{-except_pg_nets}- applies routing blockage to all the nets except PG nets

14) create_place_blockage

Creates a placement blockage for the specified area.

-rects {x1 y1 x2 y2} -area of the blockage

-all_macros -keeps the placement blockage around all macro

-type -hard/soft/{-partial -density -exclude_flops}/macro only

a) hard -neither standard cell nor macros may be placed in the blockage

b) soft - allows only buffers, inverters, isolation cells, clock gates, tie cells and level shifters.

c) partial -creates a placement blockage that has a maximum density as specified (sometime excludes flops and latches too)

d) macro_only - enable proto_design command to keep macros out of the placement blockage

-inst instance name - specifies the name of instances in which placement blockage has to be applied.

15) add_endcaps

Places physical only end cap cells at the end of site rows. add_endcaps is always done after floorplan and before add_well_taps

{-rect x1 y1 x2 y2}— boundary box.

{-core_boundary_only}—specifies that endcaps cells are placed within the core boundary.

{-power_domain power domain name }—specifies the power domain name in which the endcaps are inserted.

16) add_well_taps

Add physical only well tap cells. Well tap cells are physical only filler cells that are required by some technology libraries to limit the resistance between power and ground connections.

{-area x1 y1 x2 y2}-specifies the co ordinates of all row in which well tap needed to be placed

{-cell_interval microns } -specifies the maximum distance from the center of one well tap to the other well tap

{-checker_board} —places the well tap in checker board pattern

17) add_decaps

Adds decoupling capacitance to the end of design.

[-place_status] -keep the place status of macro as fixed or placed

[-area lllx lly urx ury | -exclude _areas { {lax lly urx ury} ... { } }] -defines the entire area in which decaps are to be added

[-pg_net net name] —add decaps to the specified power rail.

18) check_floorplan -report_density

Checks the quality of the floorplan to detect potential problems before the design is passed on to other tools.

{-outfile filename} - outputs detailed information for the specified blocks.

{-report_density} - Reports target utilization((TU) and effective utilization (EU) for the entire design, fences and regions.

19) create_place_halo

Adds a halo to the block. A halo is an area that prevents the placement of standard cells within the specified halo distance from the edge of a hard macro to avoid congestion.

{-all_blocks} — add halo around all hard macros, black boxes and committed partitions.

{-all_io_pads} —add halo around all IO Pads.

{-cell name } —Add halo around all instances of cell.

Sanity check after Powerplan:

1) check_connectivity

Detects conditions such as opens, unconnected, wires(geometric antennas), unconnected pins, loops, partial routing and unrouted nets and generates violation marker in the design window; reports violations.

{-out_file filename } specifies the report file for connectivity violation data

2)check_power_vias

This command checks missing powergrid vias.

{-area {x1 y1 x2 y2}} specifies the co-ordinates of the area to be checked

{-report filename } specifies the name of output file for the report.

{-what_if_report filename } write out the violations data into an ECO file that can be used directly as an input for what if rail analysis

3) check_drc

Checks for the DRC Violations and creates violation markers in the design database that can be seen on the GUI and browsed with the violation browser.

{-area {x1 y1 x2 y2} ... {}} checks DRC within the specified area

{-out_file filename } specifies the report file that contains DRC violations information.

4) add_rings -around {each_block} -layer {layer1 layer2 } -offset {a b c d} -spacing {value} -width {value}

Create rings for the specified nets around the core boundary or selected blocks.

5) add_stripes -area

Creates power strips within the specified area.

6) update_power_vias

Adds power vias to the design or perform one of the following actions to existing power vias:

a) Modifies the power vias

b) Deletes the power vias

c) Fixes the vias that violate LEF minimum cut rule

INNOVUS command Placement:

1) set_cell_padding

Add cell padding to the specified instance

{-cells leaf cell name }

{-padding right | left | top |bottom}

2) set_inst_padding

creates cell padding for entire hierarchy of instances.

{-inst instance name }

{-padding number of sites [-right_side numofsites] [-left_side

numofsites] [-top_side numofsites] [-bottom_side numofsites]

3)create_group

Command for creating fence, region, guide, cluster

{-name group_name } Name of the created group

{-type fence/region/guide/cluster} specifies the type of group to create

{-density value } specifies the placement density percentage

{-rects {x1 y1 x2 y2}.....{ } }specifies the coordinates of the rectangle for fence, region and guide

4)group_path

Creates group path in the design, and identifies them with the path group name.

{-name path group name } specifies the name of path group

{-from fromlist } List of pins, instances at the start of path

{-through throughlist } list of pins, instances where the paths go through

{-to to list } list of pins, instances where the path ends

5) place_connected

Magnetic placement

{-attractor_pin pinlist } specifies hard macros/I/Os/fixed standard cells for attracting standard cells.

{-attractor macrolist } places only standard cells connected with specified pins of attractor close to attractor.

{-sequential all_connected } pulls sequential cells connected close to the attractor.

6) place_opt_design

Executes pre-CTS with both placement and pre CTS optimization.

{-expanded_views} prints timing information for each active view at the end of the command.

{-incremental} specifies the “place_opt_design” to run incremental mode

{-num_paths numofpaths } specifies the number of paths to be reported in the timing report.

{-report_dir directoryname } specifies the output directory in which the timing/DRV reports will be saved.

{report_prefix outfilename } specifies the file name that is generated at the end of place_opt_design.

NOTE - place_opt_design is made from two commands which were used separately place_design and opt_design -pre_cts.

7) opt_design

Performs timing optimization before or after the clock tree is built, or after routing and generates timing reports.

{-drv} — corrects max_cap and max_trans violations, if we want to correct fanout_load we have to first specify opt_fix_load attribute before you specify opt_design.

{-expanded_views} generates detailed view-specific timing reports at the reports at the ends of opt_design command.

{-hold} corrects hold violations.

{-hold_vioaltion_report filename } generates the following report files for the remaining violation that remain after hold fixing.

{-incremental} performs setup optimization incrementally on the violated

paths until they cannot be optimized more. Both WNS and TNS are optimized.

{-post_cts} performs timing optimization o design whose clock tree has been created.

{-post_route} performs timing optimaztion on the design whose routing is complete.

{-pre_cts} performs timing optimization on the place design, before the clock tree is built.

{-report_dir directoryname **}** reports are generated in the given directory.

{-setup} corrects setup violations for the post_route optimization flow. When -hold parameter is given, a combined setup and hold optimization is performed.

8) report_congestion

Reports the average congestion and the local hotspot score.

{-hotspot} reports the local hotspot score. It reports the local and total hotspot area.

{-overflow} reports horizontal and vertical overflow.

{-3d} honors 3d congestion map.

9) set_db opt_multi_bit_flop_opt true

To enable multi bit flip flop conversion

10) set_db opt_power_effort medium/high

To enable leakage power optimization.

INNOVUS Commands CTS stage:

1) ccopt_design

Performs clock concurrent optimization (CCOpt) on the current loaded design in Innovus.

{-check _cts_config} checks that all the prerequisites for running the clock tree synthesis are fulfilled without actual doing CTS.

2) create_clock_tree_spec

Create a clock tree network with associated skew groups and other clock tree synthesis (CTS) configuration setting such as ignore pins, case analysis, max trans etc.

{-out_file filename } writes this clock tree specification script file in Stylus UI format.

{-views} specifies the TCL list of analysis view names.

3) delete_clock_tree_spec

This command deletes all the skew group definitions and other clock tree synthesis configuration information.

INNOVUS command Routing Stage:

1) route_design

Runs routing per postroute via or wire optimization using the nanoroute router. Using without any arguments runs global and detailed routing.

{-placement_check} -checks for placement violation before routing.

{-track_opt} perform timing optimization after track assignment.

{-via_opt} optimizes vias after routing

2) opt_design -post_route

Performs timing optimization after post route.

3) time_design -report_dir filename

Runs early global route, extraction and timing analysis, and generates detailed timing reports. The generated timing reports are saved in the directory that you specified using -report_dir

PrimeTime Commands:

1) **{-from / -rise_from / -fall_from fromlist }**
-from

Reports only paths that start from the specified list of pins, ports, nets, to cell instances; po from start point clocked by the named clocks.

-rise_from

Same as -from option except that the path must start with rising transition

-fall_from

Same as -rise_from option, except that the path must start with a falling transition.

2) **{-to / -rise_to / -fall_to }**

Reports only paths that end at the specified list of pins, ports, nets, or cell instance; or at endpoints clocked by the named clocks.

3) **{-exclude excludelist }**

Excludes all paths in which the data goes from, through, or to the specified list of pins, ports, nets or cell instances. Use of -exclude option can lead to longer runtimes. To reduce the runtime impact, use -from and -through.

4) **{-through / -rise_through / -fall_through }**

For ex. **report_timing -from -A1 -through {B1 B2} -through C1 -to D1**

Reports only paths in which the data goes through the named pins, ports, cell instances, or nets. This option can be used multiple times in a command.

If advanced analysis through transparent latches is enabled (timing_enable_through_paths set to true), and the pins specified in the last -through options are latch loop breakers.

5) {-delay_type delaytype }

Specifies the type of path delay constraint to consider for finding and sorting paths with worst slacks.

-max - max delay (setup constraint)

This option is considered as the default option.

-min - hold constraint

-min_max - both min and max delay

6) {-nworst paths per endpoint }

Reports up to the specified number of worst path per endpoint. By default the value is 1.

If you are setting the value greater than the default value, then the tool does the following things:

a) Implicitly sets -max_paths equal to the -nworst setting if the max_path option is not used in the command, so that at least one set of multiple paths to an endpoint can be reported.

b) Implicitly sets -slack_lesser_than 0.0 if the slack_lesser than and -slack_greater_than options are not used in the command, so that only violating paths are violated.

7) {-max_paths max path count }

Reports up to the specified maximum total number of paths.

If -group option is specified, the total number of paths reported is at most the specified number of paths per path group.

If -group is not specified, the total number of paths reported is at most the specified number of paths among all path group.

The default max_paths_count is equal to the -nworst setting, or the -slack_lesser_than and -slack_greater_than options are not specified.

Difference between -max_paths and -nworst:

- The -nworst option controls the maximum number of paths reported per endpoint.
- -max_paths options controls the overall total maximum number of paths reported.
- The -max_paths setting should be at least as larger than the -nworst setting, so by default the tool implicitly sets -max_paths to the same value as -nworst setting, so by default the tool implicitly sets -max_paths to the same value as -nworst if the -max_paths option is not specified.

8) {-group groupname }

Reports only paths that belongs to the specified path groups.

Paths groups are created by create_clock and group_path commands. Without -group command the tool will report the WNS of the whole design.

report_timing -group [get_path_groups]

Reports the single worst path in each path group.

9) {-slack_lesser_than maximumslack }

Reports only paths with slack less than the specified maximum_slack value

10) {-slack_greater_than minimumslack }

Reports only commands with slack greater than the minimum_slack value.

11) {-pba_mode none / path / exhaustive / ml_exhaustive}

Specifies one of the following path-based timing analysis reports

-none(default)

Disables path-based analysis and enables ordinary graph-based analysis. This is the fastest mode.

-path

Performs path based analysis on paths after they have been gathered by GBA, performing more accurate timing results for those paths.

-exhaustive

Performs an exhaustive path_based analysis to determine the truly worst case paths in the design. This can be used along with

-start_end_pair.

-ml_exhaustive

Performs Machine learning based exhaustive path-based analysis.

12) {-start_end_type from_to_type }

Restricts the report to one of four classes of paths based on the type of startpoint and endpoint.

reg_to_reg - register to register

reg_to_out - register to output port

In_to_reg - input to register

In_to_out - Input port to Output port

13) {-don_merge_duplicates}

Prevents the merging of duplicate paths across multiple scenarios in DMSA.

By default, when the same path is reported in more than one scenario, the tool reports only the single most critical instances of the path from all scenarios.

14) {-path_type format}

Specifies the type of path information displayed in the timing report, using one of the following format keywords:

-summary

Displays the path startpoint, path endpoint, and slack.

-end

Displays the path endpoint, path delay, required time, and slack.

-short

Displays a report like the default full report but omits the intermediate points between the startpoint and endpoint.

-full (default)

Displays the full data path, including all intermediate points and their incremental points between start and endpoint.

-full_clock

Displays a full report with the addition of the clock source to the launch and capture points.

-full_clock_explained

Displays a full_clock report with the addition of the clock path points between primary clock source and its related generated clock source point.

15) {-nets}

Shows the nets in the timing path.

16) {-transition_time}

Shows the transition time (slew) in the path report for each driver and load pin, appearing as an additional column labeled “Trans”.

17) {-crosstalk_delta}

Reports the annotated delta delay values at cell input pins in a column labeled “Delta”. The delta delay values are computed during crosstalk and Signal Integrity analysis.

18) {-derate}

Shows derating factors in a column labeled “Derate”. Specifying this option automatically sets the -input_pins option, so that different net and cell derating values are reported.

19) {-variation}

Includes parametric on-chip variation (POCV) in the report in columns labeled “Mean” and “Sensit”. POCV analysis is enabled by setting the **timing_pocvm_enable_analysis** variable to true.

20) {-physical}

Reports the X-Y co-ordinates for each path element in a column labeled “location”.

21) {-voltage}

Reports the supply group or the supply net name for each path element in a column labeled “voltage” allowing you to debug voltage levels in multi voltage designs.

22) {-sort_by slack / group}

If there is nothing the default option is slack. If “group” option is enabled the paths are sorted according to the group.